MIL-C-21852E
INT. AMENDMENT-2(GL)
3 June 1988
SUPERSEDING
INT. AMENDMENT-1(GL)
4 August 1986

MILITARY SPECIFICATION

CLOTH, TAFFETA, NYLON

This interim amendment is approved for use within the Department of the Army, with MIL-C-21852E dated 25 August 1981.

PAGE 1

1.2, under "Type III": Add the following:

"Class 1 - Plain finish
Class 2 - Water repellent treated"

PAGE 2

2.2: Delete "(Elemendox)" and add "(Elemendorf)".

PAGE 3

Table I, in "Type IV" column for "Yarns per inch (min)" warp and filling: Delete "108 (43.2)" and "71 (28.4)" and substitute: "100 (39.4)" and "68 (26.8)" respectively.

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3.5: Delete" and substitute:

"3.5 Color. The color of the dyed cloth shall be Camouflage Green 483 (see 6.7) or as specified (see 6.2) and shall match the standard sample."

3.5.1: Delete" and substitute:

"3.5.1 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescant lamplight at 2300 ± 200 K."

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Add the following:

"3.5.2.3 Colorfastness type III class 2 only. The finished cloth shall show fastness to light, dry cleaning, laundering and perspiration equal to or better than the standard sample or equal to or better than a rating of "good". The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of not lower than 3.5. Testing shall be as specified in 4.5.

3.5.3 Spectral Reflectance. Cloth dyed Camouflage Green 483 shall conform to the reflectance values specified below when tested as specified in 4.5.4.

Spectral Reflectance Requirements

Wavelength Nanometers (nm)	Reflectance (%) Max.	Min.
600	10	3
620	10	3
640	10	3
660	11	3
680	13	3
700	28	4
720	40	5
740	52	7
760	60	11
780	64	17
800	67	24
820	70	32
840	71	37
860	73	40"

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3.7.3, Line 1: Delete "nonfibrous materials" and substitute "nonfibrous materials (class 1 only)."

Add the following:

"3.7.5 Water repellency (Type III, Class 2 only). The cloth shall be given an approved fluorocarbon type water repellent treatment (see 6.6) and shall conform to the requirements of 3.7.5.1 through 3.7.5.3 when tested as specified in 4.5.

3.7.5.1 Dynamic absorption. The dynamic absorption initially and after 3 launderings shall be not more than 8 percent for the average of the sample units with no sample unit greater than 10 percent.

3.7.5.2 Spray rating. The results of the three individual determination on the sample unit for spray rating shall be equal to or better than ratings 100, 100, 90 initially and 80, 80, 70 after 3 launderings.

3.7.6 Resistance to organic liquids (Type III, Class 2 only). The finished cloth shall show no wetting by n-tetradecane initially and after 3 launderings when tested as specified in 4.5."

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Table II: Add the following:

"Characteristic	Requirement Paragraph	Test method
Spectral reflectance	3.5.3	4.5.4
Water repellency (Type III, Class 2 only)	3.7.5	6/
Dynamic absorption		
Initial	3.7.5.1	5500
After 3 launderings	3.7.5.1	5552, 5500, 7/
Spray rating		
Initial	3.7.5.2	5526
After 3 launderings	3.7.5.2	5552, 5526, 7/
Resistance to organic liquid		
Initial	3.7.6	4.5.3
After 3 launderings	3.7.6	5552, 4.5.3, 7/"

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Table II, add the following:

"6/ The contractor shall report the water repellents used.

7/ Specimens shall be subjected to 3 complete cycles (wash and dry) prior to determination of hydrostatic pressure, dynamic absorption, spray rating, and resistance to organic liquid after laundering."

PAGE 10

Add the following:

"4.5.3 Resistance to organic liquid test. Place a small specimen of the cloth face up on a smooth horizontal surface. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After one minute, examine the specimen under light at an angle. Absence of light reflectance at the fabric-drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample shall be tested. Evidence of wetting on one or more specimens shall constitute a test failure.

4.5.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600-860 nm relative to a barium sulfate standard, the preferred white reference standard. Other reference white standards may be used provided they are calibrated to an absolute white; i.e., Halon, magnesium oxide, or vitrolite tile. The spectral band width at 860 nm shall be less than 26 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simultes either CIE Source A or CIE Source D65 in the visible spectrum. The specular component of reflection shall be included in the measurement. Specimens shall be measured as a single layer of fabric backed by seven layers of the same fabric. Readings will be taken on a minimum of two different areas and the data averaged. The specimen shall be viewed at an angle no greater than 10 degrees from normal. Photometric accuracy of the spectrophotometer shall be within 1 percent; and the wavelength accuracy shall be within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. Failure of the near-infrared spectral reflectance requirements will occur when spectral reflectance limits are exceeded at four or more specified wavelengths."

Para. 6.1: Add to Type III uses, "Sleeping bags (Army)".

6.2 b., after "Type": Add "and class".

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Add the following:

"6.6 Approval of fluorocarbon water repellent. Approval of the fluorocarbon type water repellent treatment is the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 and is based on extensive tests, including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation (approximately 6 months), only those chemical treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

6.7 Dye Formulation for Camouflage Green 483. A suggested but not mandatory dye formulation for Camouflage Green 483 is as follows:

Acid Orange 162
Acid Blue 171

6.8 White standards. Barium sulfate of suitable quality for use as a white standard is available from Eastman Kodak Company. Tiles are available from the instrument manufacturer (see 4.5.4).

6.9 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the visible/near-infrared as the Diano Hardy, Diano Match Scan, Hunter D54P-IR, Hunter VIS/NIR Spectrocolorimeter, and Macbeth 1500 with IR option."

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

Project No. 8305-A272

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